ANSWER KEY & MARKING SCHEME SET 1

Chapter 5: **Describing Motion Around Us** | Class 9 Science | Maximum Marks: 40

SECTION — A (Objective Type) — Answers — 10 × 1 = 10

Q.	Correct Option	Brief Reasoning
1	**(a)** Both A and R are true and R is the correct explanation of A.	R explains A.
2	**(b)** u / sqrt(2)	v squared = u squared - 2g(h/2) with h = u squared/2g.
3	**(a)** 8 km/h	24 km/ 3 h.
4	**(a)** Both A and R are true and R is the correct explanation of A.	R explains A.
5	**(a)** 1.5 t	v = at; retard time = at/2a = t/2.
6	**(c)** 4.8 km/h	Harmonic mean.
7	**(a)** Both A and R are true and R is the correct explanation of A.	R explains A.
8	**(c)** 2t	t proportional to sqrt(h).
9	**(a)** Both A and R are true and R is the correct explanation of A.	If a body returns to its starting point, the shortest distance between start and finish is zero, although it has travelled a path.
10	**(b)** Both A and R are true but R is not the correct explanation of A.	Both are true statements about graphs, but the second does not explain the first.

SECTION — B (Very Short Answer) — Answers — 4 × 2 = 8

Q.	Expected Answer / Value Points	Marks
11	Harmonic mean 2 x 30 x 60/90 (1); = 40 km/h (1).	1+1
12	Dropped: 2 s (1); Thrown down: 20 = 10t + 5t squared gives t = 1.24 s (1).	1+1
13	a = v squared / 2s (1); = 225/120 = 1.875 m/s squared (1).	1+1
14	u = g x 2 = 20 m/s (1); H = 20 m (1).	1+1

SECTION — C (Short Answer) — Answers — 4 × 3 = 12

Q.	Expected Answer / Value Points	Marks
15	v = 20 m/s; s1 = 50 m (1); s2 = 200 m (1); t3 = 10 s; s3 = 100 m; total = 350 m (1).	1+1+1
16	s1 = a/2 = 2, so a = 4 m/s squared (1); s4 = 0.5 x 4 x 16 (1); = 32 m (1).	1+1+1
17	v = 8 m/s; s1 = 16 m (1); s2 = 48 m (1); t3 = 2 s; s3 = 8 m; total 72 m (1).	1+1+1
	OR: Relative speed = 18 km/h = 5 m/s (1); Relative distance = 100 m (1); Time = 20 s (1).	1+1+1
18	t = 1 s, v = 10 m/s (1); a = 10/0.1 = 100 m/s squared (1); Bending knees increases the stopping time (1).	1+1+1
	OR: Closing speed with truck 45 m/s (1); In 6 s: 270 m covered (1); 270 < 400, so safe (1).	1+1+1

SECTION — D (Long Answer) — Answers — 2 × 5 = 10

Q.	Expected Answer / Value Points	Marks
19	(i) $v = u + at = 0 + 0.5 \times 40 = 20$ m/s (1). (ii) $s = \frac{1}{2}at^2 = \frac{1}{2} \times 0.5 \times 1600 = 400$ m (1). (iii) $s = vt = 20 \times 60 = 1200$ m (1). (iv) Total $= 400 + 1200 = 1600$ m (1); average speed $= 1600/100 = 16$ m/s (1).	1+1+1+2
20	Initial velocity 10 m/s upward (1); -75 = 10t - 5t squared (1); t squared - 2t - 15 = 0, t = 5 s (1); v = 10 - 50 = -40 m/s (1); Speed 40 m/s downward (1).	1+1+1+1+1
	OR: Equal distances: t squared = 20t (1); t = 20 s (1); Distance 400 m (1); Police speed 40 m/s (1); Relative speeds before and after (1).	1+1+1+1+1

Mistakes to avoid:

- Using distance instead of displacement (or speed instead of velocity); for a round trip the displacement and average velocity are zero.
- Not converting km/h to m/s before using the equations of motion, or mixing km with seconds.
- Putting a positive sign for retardation in $v = u + at$ or $v^2 = u^2 + 2as$; for a slowing body a is negative.
- Reading the slope of an x-t graph as acceleration and the slope of a v-t graph as velocity; slope of x-t is velocity, slope of v-t is acceleration.

- Calculating the distance from a v-t graph by adding signed areas or by taking only one segment; add the areas of all segments (triangles and rectangles) and take area below the axis as positive for distance.
- Thinking that acceleration is zero at the top of a vertical throw; only the velocity is zero, acceleration is still g.

Evaluator's note: Award step marks as shown; accept any correct alternative method. In internal-choice questions, evaluate only the option attempted. Deduct up to ½ mark for missing units or figures where required.

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ANSWER KEY & MARKING SCHEME SET 2

Chapter 5: **Describing Motion Around Us** | Class 9 Science | Maximum Marks: 40

SECTION — A (Objective Type) — Answers — 10 × 1 = 10

Q.	Correct Option	Brief Reasoning
1	**(d)** 1 : 3 : 5	Odd numbers.
2	**(c)** 31.25 m	t up = 2.5; h = 0.5 x 10 x 6.25.
3	**(a)** Both A and R are true and R is the correct explanation of A.	R explains A.
4	**(c)** 12 m/s squared	s3 = a/2 x 5 = 30.
5	**(a)** 100 m	Average 20 x 5.
6	**(a)** Both A and R are true and R is the correct explanation of A.	R explains A.
7	**(b)** 100 m	10 x 20 = 200 total; minus 100.
8	**(d)** 2.8 m/s squared	Change = 14.1 m/s over 5 s.
9	**(c)** A is true but R is false.	A is the definition. R is false: for a slowing body (retardation) the acceleration is opposite to the motion.
10	**(d)** A is false but R is true.	A speedometer gives only the magnitude (speed), so A is false; R is true.

SECTION — B (Very Short Answer) — Answers — 4 × 2 = 8

Q.	Expected Answer / Value Points	Marks
11	a = (121 - 36)/68 = 1.25 m/s squared (1); t = 4 s (1).	1+1
12	Uniform circular motion (1); Satellite or a car taking a circular turn (1).	1+1
13	Initial velocity 2 m/s, a = 6 m/s squared (1); v = 2 + 12 = 14 m/s (1).	1+1
14	Displacement (diameter) is less than the arc length (1); Velocity is displacement/time (1).	1+1

SECTION — C (Short Answer) — Answers — 4 × 3 = 12

Q.	Expected Answer / Value Points	Marks
15	h = 20 x 1 - 5 = 15 m (1); v = 20 - 10 = 10 m/s upward (1); Ball is still rising (1).	1+1+1
16	u = 10 m/s (1); a = -4 m/s squared (1); Stops at t = 2.5 s (1).	1+1+1
17	First 20 m forward (1); Next 20 m backward (1); Displacement 0; distance 40 m (1).	1+1+1
	OR: v = 30 m/s = 108 km/h (1); s = 900 m (1); Using time of 60 s (1).	1+1+1
18	Circumference = 62.8 cm (1); Speed = 62.8 cm/min (1); Displacement = 20 cm (1).	1+1+1
	OR: t = (v - u)/a (1); Substitute into s = ut + 0.5 a t squared (1); v squared = u squared + 2as (1).	1+1+1

SECTION — D (Long Answer) — Answers — 2 × 5 = 10

Q.	Expected Answer / Value Points	Marks
19	(i) $72 \times \frac{5}{18} = 20$ m/s (1). (ii) $t = \frac{v-u}{a} = \frac{0-20}{-5} = 4$ s (1). (iii) $s = \frac{v^2-u^2}{2a} = \frac{0-400}{-10} = 40$ m (1). (iv) $v^2 = u^2 + 2as = 400 - 2 \times 5 \times 30 = 100$ (1), so $v = 10$ m/s (the car hits the barrier) (1).	1+1+1+2
20	Stone 1: h = 20t - 5t squared (1); Stone 2: h = 20(t - 2) - 5(t - 2) squared (1); Equate: t = 3 s (1); h = 15 m (1); Velocities 10 up and -10 down (1).	1+1+1+1+1
	OR: Area 0-2: 6 m (1); Area 2-5: 18 m (1); 5 to 6.5 s: +4.5 m, 6.5 to 7: -0.5 m (1); Displacement 28 m, distance 29 m (1); Average acceleration = (-2 - 0)/7 = -0.29 m/s squared (1).	1+1+1+1+1

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- Putting a positive sign for retardation in $v = u + at$ or $v^2 = u^2 + 2as$; for a slowing body a is negative.
- Reading the slope of an x-t graph as acceleration and the slope of a v-t graph as velocity; slope of x-t is velocity, slope of v-t is acceleration.

- Calculating the distance from a v-t graph by adding signed areas or by taking only one segment; add the areas of all segments (triangles and rectangles) and take area below the axis as positive for distance.
- Thinking that acceleration is zero at the top of a vertical throw; only the velocity is zero, acceleration is still g.

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ANSWER KEY & MARKING SCHEME SET 3

Chapter 5: **Describing Motion Around Us** | Class 9 Science | Maximum Marks: 40

SECTION — A (Objective Type) — Answers — 10 × 1 = 10

Q.	Correct Option	Brief Reasoning
1	**(a)** 1 : 2	v proportional to sqrt(h).
2	**(a)** Both A and R are true and R is the correct explanation of A.	R explains A.
3	**(b)** 10 m/s squared	60 - 20 = a x t squared = 4a, so a = 10.
4	**(d)** 100 m	0.5 x 10 x 20.
5	**(d)** A is false but R is true.	A is false and R is true.
6	**(c)** 4 m/s squared	Compare with ut + (1/2)at squared.
7	**(a)** 625 m	125 + 500.
8	**(a)** Both A and R are true and R is the correct explanation of A.	R explains A.
9	**(a)** Both A and R are true and R is the correct explanation of A.	Gravity gives a constant downward acceleration of about 10 m/s^2 even when the velocity momentarily becomes zero.
10	**(b)** Both A and R are true but R is not the correct explanation of A.	Both are true but unrelated.

SECTION — B (Very Short Answer) — Answers — 4 × 2 = 8

Q.	Expected Answer / Value Points	Marks
11	v squared = 225 + 400 (1); v = 25 m/s (1).	1+1
12	Velocity increases with time (1); Equal times, increasing average velocity (1).	1+1
13	t proportional to sqrt(h): 1 : 2 (1); v proportional to sqrt(h): 1 : 2 (1).	1+1
14	Symmetry of motion (1); The first second up equals the last second down (1).	1+1

SECTION — C (Short Answer) — Answers — 4 × 3 = 12

Q.	Expected Answer / Value Points	Marks
15	a = 2 m/s squared (1); s = 8 x 6 = 48 m (1); v at 3 s = 8 m/s (1).	1+1+1
16	Total time 4.47 s (1); First 50 m takes 3.16 s (1); Second 50 m takes 1.31 s as speed has increased (1).	1+1+1
17	Energy or motion is symmetric (1); Upward and downward trips are mirror images (1); Speed equals initial speed (1).	1+1+1
	OR: Speeds 2 m/s and 5 m/s (1); Distances 20 m and 50 m (1); Separation 30 m (same direction) (1).	1+1+1
18	a = 3 m/s squared (1); s = 37.5 m (1); Time = 5/1.5 = 3.33 s (1).	1+1+1
	OR: Reaction distance 8 m (1); Braking distance 40 m (1); Total 48 m which exceeds 40 m: danger (1).	1+1+1

SECTION — D (Long Answer) — Answers — 2 × 5 = 10

Q.	Expected Answer / Value Points	Marks
19	(i) $a = \frac{8-0}{4} = 2$ m/s^2 (1). (ii) Uniform velocity, with zero acceleration (1). (iii) $\frac{8-0}{2} = 4$ m/s^2 (1). (iv) Area $= \frac{1}{2} \times 4 \times 8 + 6 \times 8 + \frac{1}{2} \times 2 \times 8 = 16 + 48 + 8 = 72$ m (2).	1+1+1+2
20	u = sqrt(2 x 10 x 45) = 30 m/s (1); Time up 3 s, time of flight 6 s (1); v at 2 s = 30 - 20 = 10 m/s upward (1); At 4 s: 10 m/s downward (1); Symmetry of the upward and downward motion (1).	1+1+1+1+1
	OR: At 20 m/s: 10 + 40 = 50 m (1); At 30 m/s: 15 + 90 = 105 m (1); Doubling speed more than doubles stopping distance (1); Reaction time matters (1); Speed limits are for safety (1).	1+1+1+1+1

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